

Expectation Maximization: Missing Data

Arin Ghazarian
California State University Long Beach

Missing Features

- Missing values is a common problem
- We can not use a dataset with missing values to train an ML Model
 - All values needs to be valid numerics
- Solution
 - Either delete rows which have missing values
 - Replace missing values with a good guess (imputation)
 - Mean
 - Median
 - Mode (for categorical variables)
 - Can we do better?

Python

```
np.nan
```

```
data[data.isnull()]
```

```
data[data.notnull()]
```

```
data.dropna()
```

```
df.dropna(axis='columns')
```

```
df.dropna(axis='rows', thresh=3)
```

```
data.fillna(0)
```

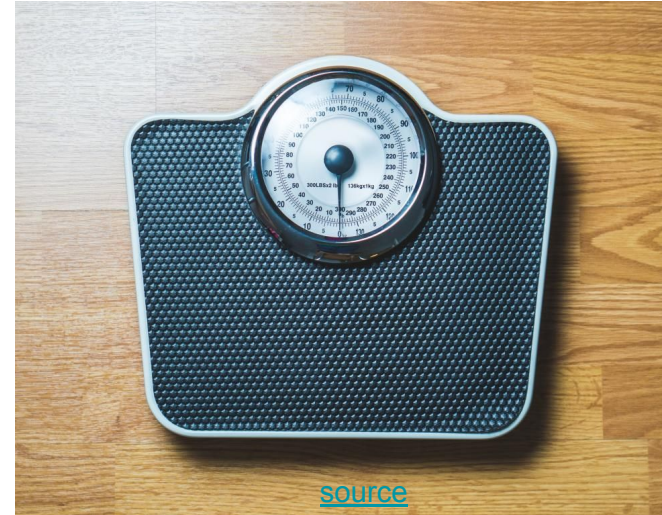
```
data.fillna(method='ffill')
```

Missing Data Models

- Every data point has some likelihood of being missing.
- The process that determines the probabilities of missing is called the missing data mechanism/response mechanism
- Three types
 - MCAR
 - MAR
 - MNAR (NMAR)

Missing Data Models

- **Missing Completely at Random (MCAR):** If the probability of being missing is the same for all cases. No specific structure can describe the missing-ness
 - An example of MCAR is a weighing scale that ran out of batteries.
- **Missing at Random (MAR):** If the probability of being missing is the same only within groups defined by the observed data. The reason a values is missing is not due to the value itself, but due to another observed variable
 - When placed on a soft surface, a weighing scale may produce more missing values than when placed on a hard surface. Such data are thus not MCAR. If, however, we know surface type and if we can assume MCAR within the type of surface, then the data are MAR.
 - People of specific gender dont like disclosing their age in a survey
- **Missing not at random (MNAR/ NMAR):** If neither MCAR nor MAR holds, then we speak of missing not at random (MNAR). The probability of being missing varies for reasons that are unknown to us. The reason a value is missing is due to the value itself.
 - The weighing scale mechanism may wear out over time, producing more missing data as time progresses, but we may fail to note this. If the heavier objects are measured later in time, then we obtain a distribution of the measurements that will be distorted.
 - MNAR includes the possibility that the scale produces more missing values for the heavier objects (as above), a situation that might be difficult to recognize and handle.
 - An example of MNAR in public opinion research occurs if those with weaker opinions respond less often.
 - In a survey about job, those who have higher incomes tend not to share their income



Latent Variables in Missing Data Problem

- Missing data/Not observed points are the latent variables. We marginalize over missing variables

$$l(\theta; X_{\text{OBS}}) = \log f(X_{\text{OBS}}; \theta) = \log \int f(X_{\text{OBS}}, X_{\text{MIS}}; \theta) dX_{\text{MIS}}.$$

- This quantity cannot be computed explicitly in general cases

Formulating Missing Data as EM

- The EM algorithm finds the MLE by iteratively maximizing the expected complete-data log-likelihood:

$$l(X; \theta) = \log f(X_{\text{OBS}}, X_{\text{MIS}}; \theta).$$

Formulating Missing Data as EM

- **E-step:** Compute the expectation of complete-data log-likelihood, with respect to the conditional distribution of missing covariate parameterized by $\theta^{(t)}$:

$$Q(\theta, \theta^{(t)}) = \mathbb{E} [l(X; \theta) | X_{\text{OBS}}; \theta^{(t)}] = \int l(X; \theta) f(X_{\text{MIS}} | X_{\text{OBS}}; \theta^{(t)}) dX_{\text{MIS}},$$

- **M-step:** Find the parameter value that maximizes the expected log-likelihood:

$$\theta^{(t+1)} \in \operatorname{argmax}_{\theta} Q(\theta, \theta^{(t)})$$

Expectation Maximization to Replace Missing Data

- Expectation maximization is applicable whenever the data are missing completely at random or missing at random-but unsuitable when the data are not missing at random
- EM is used for replacing missing data or estimating gaussian mixture model and many more
- EM is maximum-likelihood estimation in the presence of latent variables
 - Latent variables in GMMs: The identity of the component used to generate the data point
 - Latent variables in Data imputation: Missing data
- Estimates the parameters which maximize the observed data log-likelihood, then estimates the missing values.

EM Algorithm

1. Given an incomplete dataset, start with some initial parameter
2. Iterate until convergence (i.e. until parameter estimate does not change a lot between iterations)
 - a. Expectation step (E): Using the available data, estimate the missing observations
 - b. Maximization step (M): Using the complete data generated in step 2, maximize the parameters of the model

Other Resources

- [Expectation maximization to manage missing data](#)
- [EM imputation: Python implementation](#)